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1. a) Explain **extension** and what would the Database look like without it? [2]

**Ans:** Database state is called extension. It basically contains the actual data stored in database at a particular moment. Without this the database will be empty.

b) My job is to define Databases for companies. I don't take part in making the websites but, I do give them ideas on how to do it. Lastly, I am the most connected to actual users. Who am I? [1]

**Ans:** Database Designer

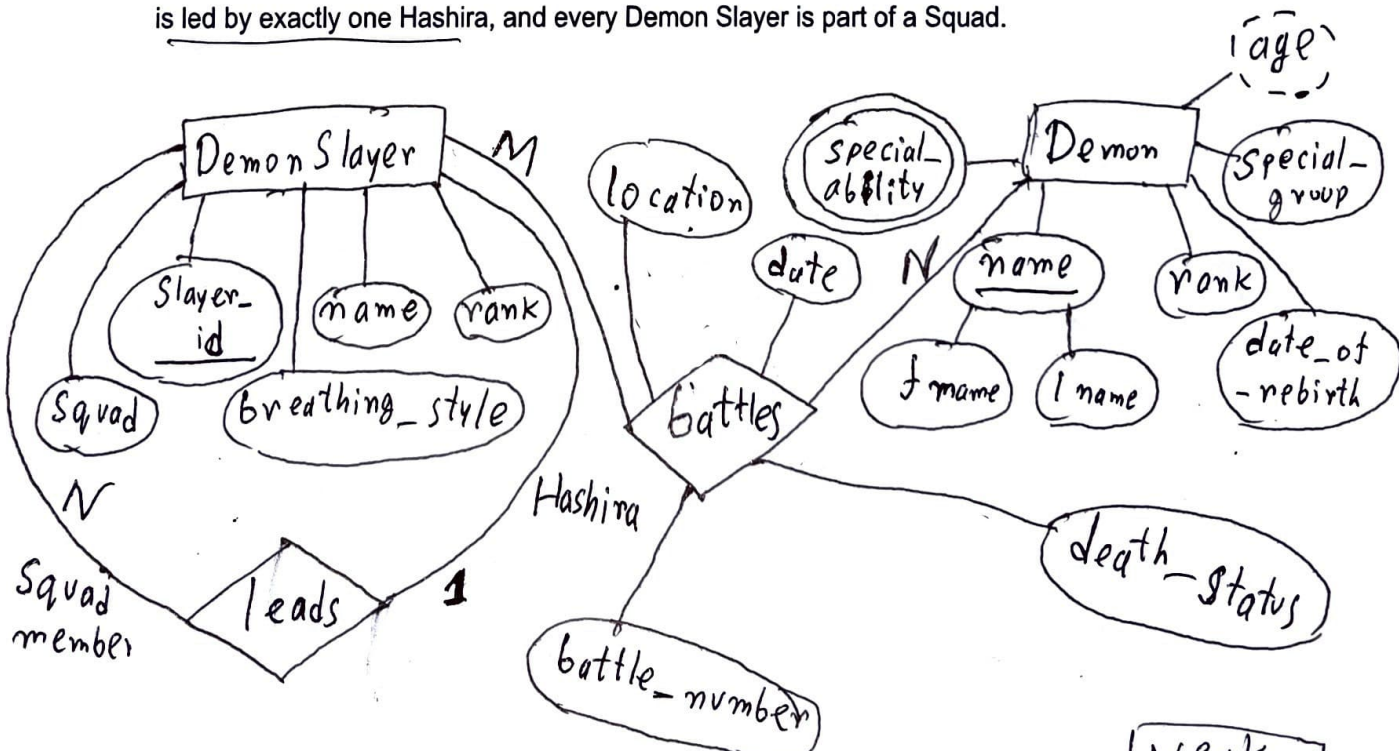
2. Demon Slayers are warriors who fight against Demons to protect humanity. Construct an ER diagram for the Demon Slayer world according to the data requirements given below: [7]

a. A **Demon Slayer** has a unique slayer\_id, a name, a rank, and a breathing\_style.

b. A **Demon** has a unique name with two parts, a rank (e.g., Lower Moon, Upper Moon), an affiliation to a special group, and one or more special\_abilities. A demon also has a date\_of\_rebirth, from which their age can be calculated.

c. **Battles** occur between Demon Slayers and Demons. Each Battle has a battle\_number (a sequential number: 1, 2, 3...), location, date. In battles Demon Slayers die and their status is recorded whether they are alive or not.

d. A Hashira is a special Demon Slayer who can lead a Squad of Demon Slayers. Each Squad is led by exactly one Hashira, and every Demon Slayer is part of a Squad.



⊛ You can also assume battle as a weak entity